NAME ……

RANDOM/PERSONAL No. ……../…… **SIGNATURE** ……

P525/1
CHEMISTRY
Paper 1
SATURDAY, 11th August 2018 (Morning)
2 hours45 minutes

ACHOLI SECONDARY SCHOOLS EXAMINATIONS COMMITTEE

Uganda Advanced Certificate of Education

Joint Mock Examinations, 2018

CHEMISTRY

Paper 1

2 hours 45 minutes

INSTRUCTIONS TO CANDIDATES:

- *Attempt**all** questions in **section A** and **six** questions in **section B**.*
- *All questions are to be answered in the spaces provided.*
- *A Periodic Table, with relative atomic masses, is supplied/attached at the end of the paper.*
- *Mathematical tables (3-figures) and non-programmable silent scientific calculators may be used.*
- *Where necessary use: Molar gas constant, $R = 8.31\ J\ mol^{-1}\ K^{-1}$*

For Examiner's Use Only																	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	**Total**

SECTION A (46 marks)

*Answer **all** questions in this section*

Question 1:

The vapour density of a mixture containing nitrogen dioxide and dinitrogentetraoxide is 38.2 at $27^{o}C$.

(a) Calculate the number of moles of nitrogen dioxide in 100g mixture. (1½ marks)

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(b) (i) Explain the term mole fraction (½ mark)

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(ii) Show that the sum of mole fraction is one for the mixture in (a) (1½ marks)

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Question 2:

Carbon monoxide undergoes disproportionation reaction.

(a) (i) Explain the term disproportionation reaction (½ mark)

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(ii) Write the equation of reaction for disproportionation reaction of carbon monoxide (1½ marks)

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(b) Using equation(s) of reaction(s) only, show that germanium(II)oxide is:

(i) basic (1½ marks)

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(ii) acidic (1½ marks)

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Question 3:

(a) State what would be observed and write equation for the reaction that would take place when:

(i) Ethanal is mixed with saturated solution of sodium hydrogen sulphite. (1½ marks)

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(ii) Methanol added to a mixture of benzoic acid and concentrated sulphuric acid and the mixture warmed. (1½ marks)

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(b) Write the mechanism for the reaction in (a)(ii) (2½ marks)

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Question 4:

A mixture of aqueous ammonia and ammonium chloride acts a buffer solution.

(a) Define the term Buffer solution (1 mark)

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(b) Describe what happens to the mixture if small amount of the following is added:

(i) Sodium hydroxide solution (1½ marks)

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(ii) Sulphuric acid (1½ marks)

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(c) Calculate the composition of alkaline buffer solution of pH = 9.75 made by mixing aqueous ammonia and ammonium chloride of both same concentration. (Base dissociation constant of ammonia solution at 25°C is $1.78\times10^{-5}\,moldm^{-3}$) (1½ marks)

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(d) State one biological application of buffer solution (½ mark)

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Question 5:

Soap can be prepared from vegetable oil.

(a) Briefly explain how vegetable oil can be extracted from a natural source (2½ marks)

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(b) Describe how soap can be prepared from a vegetable oil. State the chemical principles involved. (2½ marks)

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(c) Write equation for the reaction leading to the formation of soap (1½ marks)

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Question 6:

Both zinc and manganese are d-block elements but zinc is not regarded as a typical transition element.

(a) Explain what is meant by transition element (01 mark)

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(b) Explain why transition metals:

(i) form coloured compounds (1½ marks)

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(ii) have catalytic property (1½ marks)

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(b) To an aqueous solution of manganese (II) chloride, aqueous ethane-1,2-diammine was added drop-wise until in excess.

(i) State what was observed (01 mark)

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(ii) Write the equation (s) of reaction(s) (02 marks)

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Question 7:

Phenyl ammonium chloride undergoes hydrolysis when dissolved in water.

(a) Explain what is meant by the term hydrolysis (½ mark)

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(b) Write the expression for hydrolysis constant K_h for phenyl ammonium chloride (½ mark)

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(c) Describe briefly how a solution of phenyl ammonium chloride with pH of 3.8 can be prepared. (Base dissociation constant of phenylamine at 25°C is $1.59\times10^{-7}\,moldm^{-3}$) (2½ marks)

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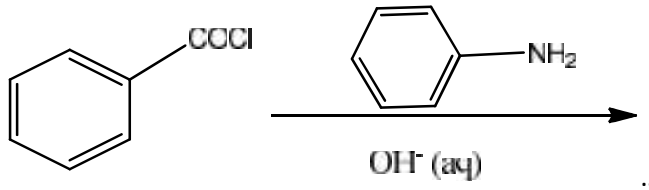
Question 8:

(a) Complete each of the following equations and in each case give the systematic name of the major organic product.

(i) $(CH_3)_2C{=}CHCH_3 \xrightarrow[\text{heat}]{MnO_4^-\,(aq)\,/\,H^+\,(aq)}$ .....(½ mark)

Name the product.....(½ mark)

(ii) 

.....(½ mark)

Name the product.....(½ mark)

(b) Write the mechanism for the reaction in (a)(ii) above (02 marks)

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Question 9:

For each of the following experiments, state what would be observed and write an equation for each of the reaction that takes place.

(a) To 1cm^3 of barium nitrate solution, 2 spatula end-fulls of zinc powder followed by excess sodium hydroxide solution were added and the resultant mixture heated.

(i) Observation (½ mark)

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(ii) Equation of reaction (1½ marks)

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(b) To 1cm^3 of silver nitrate solution, 3drops of sodium hydroxide solution are added followed by excess ammonia solution, 2-methylpropanal is added to the resultant solution and the mixture heated.

(i) Observation (½ mark)

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(ii) Equation of reaction (1½ marks)

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(c) To 1cm^3 of Iron (III) chloride solution,2-3drops of tin (II) chloride solution was added.

(i) Observation (01 mark)

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(ii) Equation of reaction (1½ marks)

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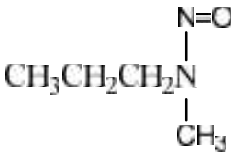
SECTION B (54 marks)

*Answer only **six** questions from this section.*

Question 10:

Write equations to show how the following compounds can be synthesised.

(a)



frompropanoic acid (03 marks)

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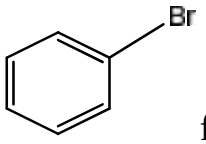
(b)



fromphenylethanoic acid (03 marks)

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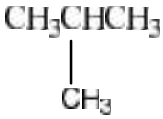
(c)



from 2-bromopropane (03 marks)

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Question 11:

A student investigated hydration of 2-methylpropene with dilute aqueous acid to form 2-methylprop-2-ol.

Step I: $CH_3C(CH_3)=CH_2 + H^+ \xrightarrow{rds} (CH_3)_3C^+$

Step II: $(CH_3)_3C^+ + H_2O \longrightarrow (CH_3)_3COH + H^+$

(a) Explain what is meant by symbol ***r.d.s*** in step I (01 mark)

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(b) Write a balanced equation of hydration of 2-methylpropene (01 mark)

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(c) Suggest the role of hydrogen ions in this mechanism. Explain your answer. (02 marks)

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(d) Use the mechanism above to suggest the rate equation for the hydration (01 mark)

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(e) Briefly explain how temperature affects:

(i) Activation energy (1½ marks)

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(ii) Rate constant (2½ marks)

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Question 12:

(a) Name two classes of hydrides (01 mark)

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(b) Write formulae of hydrides of sodium, aluminium, phosphorous, sulphur and chlorine and in each case, state type of bonding (05 marks)

Element	Sodium	Aluminium	Phosphorous	Sulphur	Chlorine
Hydride					
Type of bonding					

(c) Explain what happens when the hydride of sodium, Phosphorous and chlorine are added to water. State the nature of resultant solution and write the equation of reaction. (03 marks)

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Question 13:

The half-cell equations and their oxidation standard electrode potentials are shown below:

A $Cl^-(aq) \longrightarrow 1/2\ Cl_2(g) + e^-$ $E^o = -1.36V$

B $1/2\ H_2(g) \longrightarrow H^+(g)$ $E^o = 0.00V$

(a) Explain what is meant by the term standard electrode potential (01 mark)

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(b) Explain the following observations:

(i) A high resistance voltmeter is used during measurement of standard electrode potential. (01 mark)

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(ii) The standard reduction electrode potential of barium is more negative than that of magnesium yet both elements belong to the same group. (1½ marks)

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(c) Draw a well labelled diagram of a feasible cell when half-cell **A** is joined to half-cell **B**

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(d) (i) Determine the e.m.f of the cell in (c) above (01 mark)

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(ii) State three physical factors that would affect the magnitude of the e.m.f of the cell

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(iii) On the cell diagram in (c) indicate the direction for the flow of electrons. Explain your answer. (½ mark)

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(e) The half-cell is shown below:

$$ClO_3^-(aq) + 6H^+(aq) + 5e \longrightarrow 1/2\ Cl_2(g) + H_2O(l) \quad E^o = +1.47V$$

What does this equation tell you about the oxidizing ability of chlorate (V) ions compared to chlorine gas? Explain your answer. (01 mark)

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Question 14:

(a) State what would be observed and write equation(s) for the reaction(s) that took place when:

(i) Ethanoyl chloride is treated with water (1½ marks)

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(ii) An aqueous solution of benzene diazonium chloride is warmed to 50^oC and treated with three drops of bromine water (2½ marks)

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(iii) To a mixture of benzene, ethanoyl chloride and anhydrous aluminium chloride at $50^{o}C$ is added few drops of a mixture of aqueous iodine and aqueous sodium hydroxide solution. (1½ marks)

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(iv) An aqueous solution of benzene diazonium chloride is treated with naphthalene-2-ol in presence of sodium hydroxide solution. (1½ marks)

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(b) Outline the mechanism for the reaction that took place in (a) (i) (1½ marks)

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(c) State one application for reaction (a) (i) in organic chemistry (½ mark)

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Question 15:

Aluminium is extracted from bauxite ore.

(a) (i) Explain what is meant by the term an ore (01 mark)

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(ii) Write the formula of bauxite ore. (½ mark)

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(b) Describe briefly how pure aluminium is obtained from pure aluminium oxide. (1½ marks)

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(c) Using equations only, show how aluminium can be converted into aluminium hydroxide. (02 marks)

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(d) Potash alum, $K_2SO_4.Al_2(SO_4)_3.24H_2O$ was dissolved in water and potassium hydrogen carbonate solution added to the resultant solution.

(i) State and explain what was observed (2½ marks)

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(ii) Write equation for the reaction that took place. (1½ marks)

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Question 16:

(a) What is meant by:

(i) State of equilibrium (½ mark)

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(ii) Dynamic equilibrium (½ mark)

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(b) State two characteristics of dynamic equilibrium (01 mark)

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(c) **1:3**mixture of nitrogen and hydrogen was passed over a certain catalyst at 450°C. At equilibrium, it was found that 2.04% by volume of ammonia was formed when the total pressure was maintained at 10 atmospheres.

(i) Calculate the equilibrium constant, Kp at 450°C for (2½ marks)

$$1/2\ N_2\,(g) + 3/2\ H_2\,(g) \rightleftharpoons NH_3\,(g)$$

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(ii) State what would happen to the equilibrium position if neon is added to the mixture in c(i) at constant pressure. (½ mark)

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(iii) Give a reason for your answer in (c) (iii) (01 mark)

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(d) The table below shows some of bond energies

Bond	H – H	N – H	N≡N
Bond energy($kJmol^{-1}$)	436	743	945

(i) Calculate the enthalpy of the reaction in (c) (i) above (1½ marks)

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(ii) State the effect on Kp when temperature is increased and explain your answer. (1½ marks)

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Question 17:

Explain each of the following observations. Illustrate your answer with equations

of reactions whereapplicable.

(a) When aqueous solution of dimethylamine is added to diammonium iron (II) sulphate solution, green precipitate insoluble in excess was formed and it rapidly turned brown. (03 marks)

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(b) The boiling points of water and bromobenzene at 760mmHg are 100 and 156°C respectively. However a mixture of the two liquids boils at 96°C at the same atmospheric pressure. (03 marks)

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(c) 2-bromo-2-methylpropane undergoes a unimolecular elimination reaction when refluxed with sodium hydroxide solution in the presence of methanol. (03 marks)

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THE PERIODIC TABLE

1	2											3	4	5	6	7	8
1.0 H 1																1.0 H 1	4.0 He 2
6.9 Li 3	9.0 Be 4											10.8 B 5	12.0 C 6	14.0 N 7	16.0 O 8	19.0 F 9	20.2 Ne 10
23.0 Na 11	24.3 Mg 12											27.0 Al 13	28.1 Si 14	31.0 P 15	32.1 S 16	35.4 Cl 17	40.0 Ar 18
39.1 K 19	40.1 Ca 20	45.0 Sc 21	47.9 Ti 22	50.9 V 23	52.0 Cr 24	54.9 Mn 25	55.8 Fe 26	58.9 Co 27	58.7 Ni 28	63.5 Cu 29	65.7 Zn 30	69.7 Ga 31	72.6 Ge 32	74.9 As 33	79.0 Se 34	79.9 Br 35	83.8 Kr 36
85.5 Rb 37	87.6 Sr 38	88.9 Y 39	91.2 Zr 40	92.9 Nb 41	95.9 Mo 42	98.9 Tc 43	101 Ru 44	103 Rh 45	106 Pd 46	108 Ag 47	112 Cd 48	115 In 49	119 Sn 50	122 Sb 51	128 Te 52	127 I 53	131 Xe 54
133 Cs 55	137 Ba 56	139 La 57	178 Hf 72	181 Ta 73	184 W 74	186 Re 75	190 Os 76	192 Ir 77	195 Pt 78	197 Au 79	201 Hg 80	204 Tl 81	207 Pb 82	209 Bi 83	209 Po 84	210 At 85	222 Rn 86
223 Fr 87	226 Ra 88	227 Ac 89															
			139 La 57	140 Ce 58	141 Pr 59	144 Nd 60	147 Pm 61	150 Sm 62	152 Eu 63	157 Gd 64	159 Tb 65	162 Dy 66	165 Ho 67	167 Er 68	169 Tm 69	173 Yb 70	175 Lu 71
			227 Ac 89	232 Th 90	231 Pa 91	238 U 92	237 Np 93	244 Pu 94	243 Am 95	247 Cm 96	247 Bk 97	251 Cf 98	254 Es 99	257 Fm 100	256 Md 101	254 No 102	260 Lw 103

THE END